KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2021
Department of Computer Science and Engineering
CSE 4115
Computer and Network Security

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Explain the terms: 'Security services', 'Cryptosystem', 'Cryptoanalysis' along with the major goals of security. Also explain a conventional cryptosystem by using a proper diagram. (13)
   b) Specify Shannon characteristics to explain good ciphers along with 'confusion' and 'diffusion'. Using a block diagram depict the symmetric cryptosystem along with the ways of the key distribution. (14)
   c) Discuss the benefits of an asymmetric cryptosystem. For RSA algorithm from $ed \equiv 1 \; mod \; \varphi(n)$ show that for a single value of e, there may exist many values of d, where symbols have their usual meanings. (08)

2. a) Define chosen-plaintext attack. Show how RSA algorithm suffers with this attack. (08)
   b) Explain the working procedure of Caesar cipher. How it is an example of substitution ciphers. (07)
   c) Discuss the properties of digital signatures. RSA algorithm can be used to implement product cipher. How? Show it? (10)
   d) Discuss the techniques, pros and cons of block and stream ciphers. What are the trends of maintaining security? Explain them. (10)

3. a) Explain the working procedure of ElGamal encryption algorithm. Why it is said to be a probabilistic encryption algorithm? Explain. (13)
   b) Explain ElGamal digital signature algorithm in detail. Also give a working example of it. (13)
   c) Discuss the major characteristics of cryptographic hash function? Why it is one-way? (09)

4. a) What is firewalls? Discuss the characteristics of various malicious code. (09)
   b) Explain AES algorithm with a block diagram. Establish a comparative study between AES and DES. (14)
   c) Discuss the characteristics of good viruses. (05)
   d) How triple DES is better than DES algorithm? Explain primitive element of Elgamal cryptosystem. (07)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is segmentation fault? Explain Buffer Overflow attack with proper code and stack diagram. (15)
   b) Show how Integer Overflow attack compromises a program's reliability and security. (15)
   c) Explain document virus using Microsoft's Dynamic Data Exchange (DDE) protocol. (05)

6. a) What are the problems associated with Segmentation and Paging? Explain how Paged Segmentation removes them. (15)
   b) Explain some countermeasures that are used against Buffer Overflow attack. (15)
   c) What is format string? (05)

7. a) Explain attacks on format string vulnerability. (15)
   b) Write short notes on the followings: (12)
      (i) DHCP Spoofing Attack. (ii) Rainbow Table Attack.
      (iii) Polymorphic Viruses. (iv) Dictionary Attack.
   c) What are covert channels? Differentiate between errors and faults. (08)

8. a) How does viruses attach with original programs? (10)
   b) Explain the working principle of Elliptic Curve Cryptography (ECC). What are the reasons to prefer ECC over RSA? (10)
   c) What is salt? How does it help against Rainbow Table attack and make password hashing more secure? (10)
   d) Explain SQL injection and web bug. (05)